PRE-ALGEBRA (8TH) · GEOMETRY

Pythagorean Squares Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Pythagorean Squares Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A right triangle has legs 3 and 4. What are the areas of the squares on the legs? Predict the area of the square on the hypotenuse.
   9 and 16; 25.
2. Does $a^2 + b^2 = c^2$ work for triangles that are not right triangles?
   No. It is true only for right triangles.

2 Explore use the tool

1. Make a right triangle with legs 6 and 8. Record the three square areas.
   36, 64 and 100.
2. Make a triangle with an angle greater than 90°. Compare $a^2 + b^2$ with c^2.
   $a^2 + b^2 < c^2$ (numbers vary).
3. Make a right triangle with no horizontal or vertical side. Use the tilted-square trick on the biggest square and show the calculation.
   Answers vary, e.g., A(1, 1), B(4, 2), C(0, 4): $10 + 10 = 20$, and $(4 + 2)^2 - 4 \times \frac{1}{2} \times 4 \times 2 = 36 - 16 = 20$.
4. Make $a^2 + b^2$ exactly 2 more than c^2. Is the angle opposite c exactly 90°?
   No. It is a little less than 90° (acute), e.g., squares 5, 10, 13 give about 81.9°.

3 Record use the tool and fill in the table

Square areas

Every row can be built on the grid. a^2 and b^2 are the two smaller squares, c^2 the biggest.

a^2	b^2	c^2	$a^2 + b^2$? c^2	Type of triangle
9	16	25	=	right
5	16	29	21 < 29	obtuse
2	5	5	7 > 5	acute
8	18	26	=	right
1	1	2	=	right ($c = \sqrt{2}$)

4 Explain what did you discover?

1. In a right triangle, $a^2 + b^2$ ___ = ___ c^2, where c is the ___ hypotenuse (longest side) ___.
2. If $a^2 + b^2 > c^2$, the triangle is ___ acute ___. If $a^2 + b^2 < c^2$, it is ___ obtuse ___.
3. To find c, take the ___ square root ___ of $a^2 + b^2$.

4. Kai says a triangle with sides 5, 6 and 8 is a right triangle because it looks close to a 6-8-10 triangle. Use squares to check.
   $5^2 + 6^2 = 25 + 36 = 61$, but $8^2 = 64$. $61 < 64$, so the triangle is obtuse, not right.

5 Apply a new situation, no tool

1. A 13-ft ladder reaches 12 ft up a wall. How far is the base of the ladder from the wall?
   5 ft
2. Find the distance between (1, 2) and (7, 10).
   $\sqrt{(36 + 64)} = 10$
3. Is a triangle with sides 9, 40 and 41 a right triangle? Show why.
   Yes: $81 + 1600 = 1681 = 41^2$.
4. A square has a diagonal of 10 cm. Find its side length, exactly and to the nearest tenth.
   $s^2 + s^2 = 100$, so $s = \sqrt{50} = 5\sqrt{2} \approx 7.1$ cm.

Teacher notes

Intended discovery: For right triangles the two leg squares exactly fill the hypotenuse square ($a^2 + b^2 = c^2$), and the converse classifies triangles: > means acute, < means obtuse.

Likely misconceptions:

- Applying $a^2 + b^2 = c^2$ to any triangle.
- Using a leg as c (c must be the longest side).
- Accepting "close" as right; on a grid $a^2 + b^2 - c^2$ is always even, so 2 is the smallest possible miss.

Answers that can vary: Explore 2, 3 and 4 and most Try this items accept many triangles.